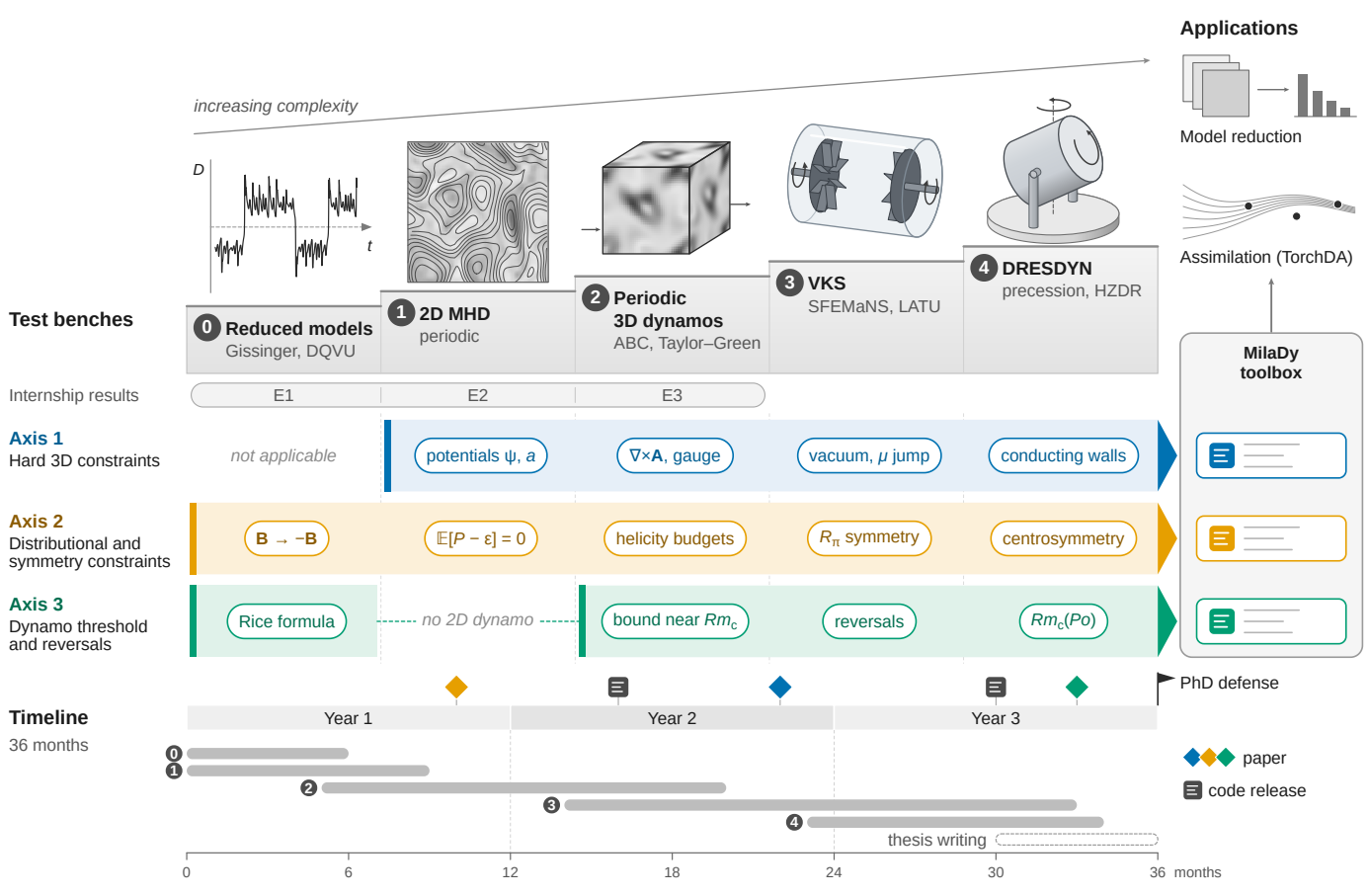

increasing complexity
D
t
Applications
Model reduction
Assimilation (TorchDA)
Test benches
0 Reduced models
Gissinger, DQVU
1 2D MHD
periodic
2 Periodic
3D dynamos
ABC, Taylor–Green
3 VKS
SFEMaNS, LATU
4 DRESDYN
precession, HZDR
MilaDy
toolbox
Internship results
E1
E2
E3
Axis 1
Hard 3D constraints
not applicable
potentials ψ, a
∇×A, gauge
vacuum, μ jump
conducting walls
Axis 2
Distributional and
symmetry constraints
B → −B
𝔼[P − ε] = 0
helicity budgets
Rπ symmetry
centrosymmetry
Axis 3
Dynamo threshold
and reversals
Rice formula
no 2D dynamo
bound near Rmc
reversals
Rmc(Po)
PhD defense
Timeline
36 months
Year 1
Year 2
Year 3
paper
code release
thesis writing
0
6
12
18
24
30
36
months